TD Learning with Function Approx

Tuesday, July 1, 2025 11:23 AM

Dataset $= \{s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}\}$ from some policy π.

At each time instant k, we obtain θ_{k+1} as a one-step gradient descent to solve

$$\min_{\theta} \frac{1}{2}\left(\phi^T(s_k,a_k)\theta - r(s_k,a_k) - \alpha\,\phi^T(s_{k+1},a_{k+1})\theta_k\right)^2 \quad \text{—} ①$$

The algorithm is motivated by the fact that

$$Q_\pi(s,a) = r(s,a) + \alpha \sum_{s'} P(s'|s,a) \sum_{a'} Q_\pi(s',a')\pi(a'|s')$$

We attempt to approximate $Q_\pi(s,a)$ as $\phi^T(s,a)\theta$, where ϕ is a known feature vector and θ is a weight vector to be learned. By the philosophy of stochastic approximation, we try to replace expectations by samples.

Writing the one-step GD of ①, we get

$$\boxed{\theta_{k+1} = \theta_k - \varepsilon_k\left(\phi^T(s_k,a_k)\theta_k - r(s_k,a_k) - \alpha\,\phi^T(s_{k+1},a_{k+1})\theta_k\right)\phi(s,a)}$$

TD-Learning

$$\Leftrightarrow \theta_{k+1} = \theta_k - \varepsilon_k \sum_{s,a,s',a'} I_{s_k=s,\,a_k=a,\,s_{k+1}=s',\,a_{k+1}=a'}\;\phi(s,a)\left(\phi^T(s,a)\theta_k - r(s,a) - \alpha\,\phi^T(s',a')\right)$$

$$= \theta_k - \varepsilon_k \sum_{s,a,s',a'} d_\pi(s)\pi(a|s)\,P(s'|s,a)\pi(a'|s')\;\phi(s,a)\left(\phi^T(s,a)\theta_k - r(s,a) - \alpha\,\phi^T(s',a')\right) - \varepsilon_k w_k$$

Expectation and the difference is w_k

As in the case of Q-learning, we will assume w_k is i.i.d. with $(s,a) \sim d_\pi(s)\pi(a|s)$

Define
$$d_\pi(s,a) = d_\pi(s)\pi(a|s)$$
$$P_\pi(s',a'|s,a) = P(s'|s,a)\pi(a'|s')$$

Let $D_\pi = \mathrm{diag}(d_\pi(s,a))$

and P is matrix with entries $P(s',a'|s,a)$

and $R =$ vector of $\bar{r}(s,a)$

$$\theta_{k+1} = \theta_k - \varepsilon_k\left(\Phi^T D_\pi \Phi\theta_k - \Phi^T D_\pi R - \alpha\,\Phi^T D_\pi P_\pi \Phi\theta_k\right) - \varepsilon_k w_k$$

where

Φ is matrix whose rows are $\phi^T(s,a)$

To proceed further, consider the following problem:

$$\min_\theta \frac{1}{2}\|\Phi\theta - Q\|^2_{D_\pi}$$

where $\|x\|^2_{D_\pi} = \sum_{s,a} d_\pi(s,a)\,x^2(s,a) = x^T D_\pi x$

In other words, we are trying to find the best approximation of the form $\Phi\theta$ to Q. It is easy to see that the problem is convex:

$$\min_\theta \frac{1}{2}\theta^T\Phi^T D_\pi\Phi - \theta^T\Phi^T D_\pi Q + \frac{1}{2}Q^T D_\pi Q$$

Differentiating w.r.t. θ:

$$\Phi^T D_\pi \Phi\theta - \Phi^T D_\pi Q = 0$$

$$\Rightarrow \theta = \left(\Phi^T D_\pi \Phi\right)^{-1}\Phi^T D_\pi Q$$

So, the best approximation to Q is given by

$$\boxed{\Phi\theta = \Phi\left(\Phi^T D_\pi\Phi\right)^{-1}\Phi^T D_\pi Q}$$

This is called the projection of Q on to the set spanned by the columns of Φ and we will denote it by

$$\mathrm{Proj}_\Phi(Q).$$

We note that the norm used for the projection $\|\cdot\|_{D_\pi}$ will be important later.

Note that TD Learning can be written as

$$\theta_{k+1} = \theta_k - \varepsilon_k\Phi^T D_\pi\left(\Phi\theta_k - T_\pi(\Phi\theta)\right) - \varepsilon_k w_k,$$

where

$$T_\pi(Q) = R - \alpha P_\pi Q$$

$$\theta_{k+1} = \theta_k - \varepsilon_k\Phi^T D_\pi\Phi\left(\theta_k - \left(\Phi^T D_\pi\Phi\right)^{-1}\Phi^T D_\pi T_\pi(\Phi\theta_k)\right) - \varepsilon_k w_k$$

$$\theta_{k+1} = \theta_k - \varepsilon_k\Phi^T D_\pi\Phi\theta_k - \varepsilon_k\Phi^T D_\pi \mathrm{Proj}_\Phi\left(T_\pi(\Phi\theta_k)\right) - \varepsilon_k w_k$$

$$= \theta_k - \varepsilon_k\Phi^T D_\pi\left(\Phi\theta_k - \mathrm{Proj}_\Phi\left(T_\pi(\Phi\theta_k)\right)\right) - \varepsilon_k w_k$$

We will now prove that $P_\Phi(T_\pi(\cdot))$ is a contraction mapping in the $\|\cdot\|_{D_\pi}$ norm.

Lemma: T_π is a contraction in the $\|\cdot\|_{D_\pi}$ norm.

Proof: $T_\pi(Q) - T_\pi(\tilde{Q})$

$$= -\alpha P_\pi Q + \alpha P_\pi\tilde{Q}$$

$$\|T_\pi(Q) - T_\pi(\tilde{Q})\|^2_{D_\pi} = \alpha^2(Q-\tilde{Q})^T P_\pi^T D_\pi P_\pi(Q-\tilde{Q})$$

$$= \alpha^2\sum_{s,a} d_\pi(s,a)\left(P_\pi(Q-\tilde{Q})\right)^2(s,a)$$

$$\leq \alpha^2\left(\sum_{s,a} d_\pi(s,a)\left[P_\pi(Q-\tilde{Q})\right](s,a)\right)^2 \quad (\text{Jensen's})$$

$$\leq \alpha^2\left(\sum_{s,a} d_\pi(s,a)\sum_{(s',a')} P_\pi(s',a'|s,a)\left(Q(s',a') - \tilde{Q}(s',a')\right)\right)^2$$

$$= \alpha^2\left(\sum_{s',a'}\underbrace{\left(\sum_{s,a} d_\pi(s,a)P_\pi(s',a'|s,a)\right)}_{d_\pi(s',a')}\left(Q(s',a') - \tilde{Q}(s,a)\right)\right)^2$$

$$= \alpha^2\|Q-\tilde{Q}\|^2_{D_\pi}$$

$\Rightarrow T_\pi$ is an α-contraction in $\|\cdot\|_{D_\pi}$ □

Claim: $\mathrm{Proj}_\Phi(T^\pi Q)$ is a contraction mapping in the $\|\cdot\|_{D^\pi}$ norm.

Proof: (a) Projection is a non-expansive operator i.e.,

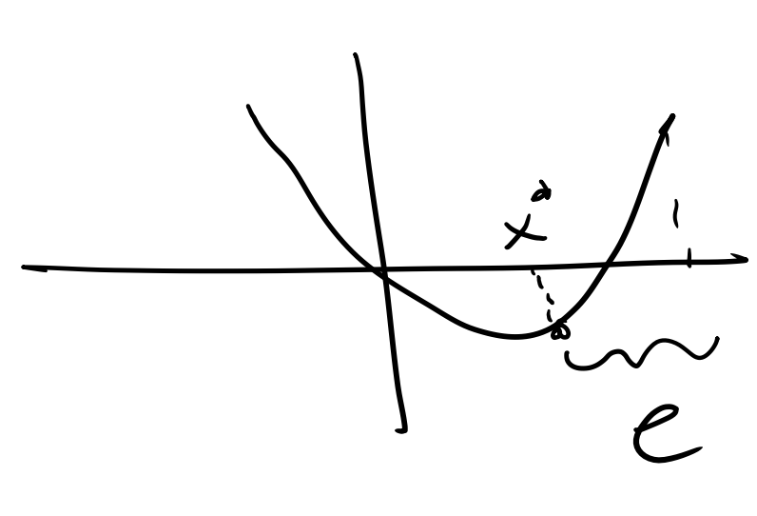
$$\|\mathrm{Proj}_\Phi(Q_1) - \mathrm{Proj}_\Phi(Q_2)\|_{D^\pi} \leq \|Q_1 - Q_2\|_{D^\pi}$$

To show this we use the following property of convex optimization:

$$\min f(x)$$
$$\text{s.t. } x \in \mathcal{C}$$

where f is a convex function and $\mathcal{C}$ is a convex set. If x^* is optimal, then

$$\langle\nabla f(x^*), x - x^*\rangle \geq 0 \quad \forall x \in \mathcal{C}$$



The projection problem

$$\min_Q \|Q - \tilde{Q}\|^2_{D^\pi}$$
$$\text{s.t. } Q \in \mathcal{C} = \{\Phi\theta : \theta \in \mathbb{R}^d\}$$

$\Rightarrow$ for the projection problem with $\tilde{Q} = Q_1$

$$\left(\mathrm{Proj}_\Phi(Q_1) - Q_1\right)^T D^\pi\left(\tilde{Q} - \mathrm{Proj}_\Phi(Q_1)\right) \geq 0 \quad \forall\tilde{Q}$$

and

with $\tilde{Q} = Q_2$,

$$\left(\mathrm{Proj}_\Phi(Q_2) - Q_2\right)^T D^\pi\left(\tilde{Q} - \mathrm{Proj}_\Phi(Q_2)\right) \geq 0 \quad \forall\tilde{Q}$$

Substitute $\tilde{Q} = \mathrm{Proj}_\Phi(Q_2)$ in the first inequality and $\tilde{Q} = \mathrm{Proj}_\Phi(Q_1)$ in the second, and add:

$$\left(\mathrm{Proj}_\Phi(Q_1) - \mathrm{Proj}_\Phi(Q_2)\right)^T D^\pi\left(\mathrm{Proj}_\Phi(Q_2) - \mathrm{Proj}_\Phi(Q_1) + Q_1 - Q_2\right) \geq 0$$

$$\|\mathrm{Proj}_\Phi(Q_1) - \mathrm{Proj}_\Phi(Q_2)\|^{\cancel{2}}_{D^\pi} \leq \left(\mathrm{Proj}_\Phi(Q_1) - \mathrm{Proj}_\Phi(Q_2)\right)^T D^\pi(Q_1 - Q_2)$$

$$\leq \cancel{\|\mathrm{Proj}_\Phi(Q_1) - \mathrm{Proj}_\Phi(Q_2)\|_{D^\pi}}\,\|Q_1 - Q_2\|_{D^\pi}$$

(b) $\|\mathrm{Proj}_\Phi(T^\pi(Q_1)) - \mathrm{Proj}_\Phi(T^\pi(Q_2))\|_{D^\pi}$

$$\underset{(a)}{\leq} \|T^\pi(Q_1) - T^\pi(Q_2)\|_{D^\pi}$$

$$\leq \gamma\|Q_1 - Q_2\|_{D^\pi}$$

□

Back to:

$$\theta_{k+1} = \theta_k - \varepsilon_k\Phi^T D_\pi\left(\Phi\theta_k - P_\Phi\left(T_\pi(\Phi\theta_k)\right)\right) - \varepsilon_k w_k$$

The proof of convergence is somewhat similar to convergence of other SA algorithms that we have seen before and will be explored in a homework assignment